

Urinary System Physiology

Integrated Biomedical and Clinical Sciences OMFS512

Shymaa Bilasy, Pharm B, Ph.D.



Mission: To advance the science and art of health through education, service, scholarship and social accountability
These teaching materials are the property of California Northstate University and may not be reproduced without permission

Learning Objectives

Learning Outcomes

1. Describe the distribution of body fluids
2. Describe the urinary system and Kidney structure
3. Describe renal functions.
4. Describe kidney function tests.

Outline

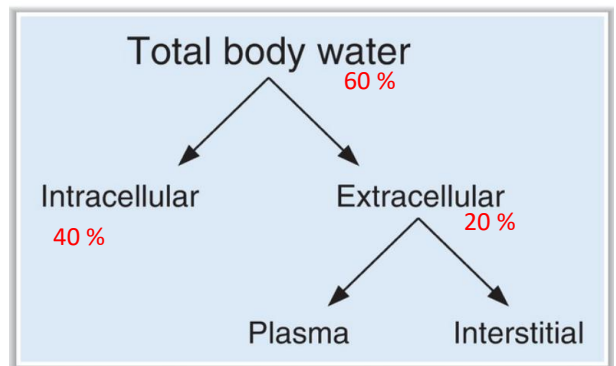
- Distribution of total body fluid.
- Structure of the Urinary System.
- Basic renal processes.
- Kidney function tests

Distribution of total body water

I) The intracellular fluid

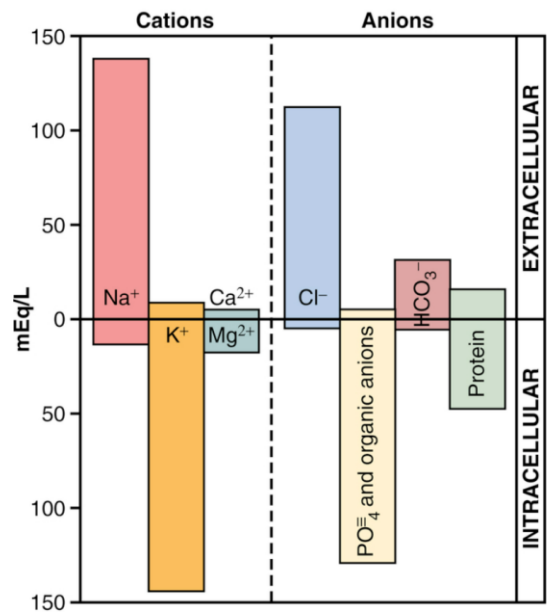
“fluid contained within the cells”.

Potassium and magnesium are the major cations of the intracellular fluid. Proteins like organic phosphate-containing molecules are the major anions.



II) The extracellular fluid: Sodium is a major cation of the ECF. Chloride and bicarbonate are major anions. ECF is the first fluid compartment to easily lose water.

- interstitial fluid: found in spaces around the cells. **Edema** refers to the accumulation of excess fluid in the interstitial space.
- Transcellular: CSF, pericardial, synovial fluid.
- Blood plasma.

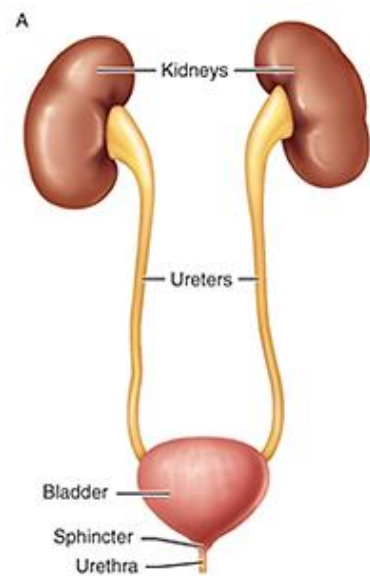


Structure of the Urinary System

California Northstate University College of Dental Medicine

6

- Kidneys are retroperitoneal organs.
- The right kidney is a bit lower and smaller than the left kidney.
- Receive 25 % of CO through the renal artery. Venous return through the renal vein.
- Produced urine is conveyed by the ureters to the urinary bladder.



Embryonic Development

- The kidneys and ureters starts developing around gestational week 4 from the mesoderm in three developmental stages (pronephros, mesonephros, and metanephros). Around week 10, the nephrons start producing urine.
- The bladder and urethra develop from the mesoderm and the endoderm.
- Horseshoe kidney is a common congenital disorder that results from the fusion of two kidneys

RELATIVELY COMMON

↳ 1 in 500

↳ twice as common in **MEN**

usually

ASYMPTOMATIC

COMPLICATIONS

↳ **HYDRONEPHROSIS** ~ urine backup

Kidney stones and increased risk of kidney cancer.



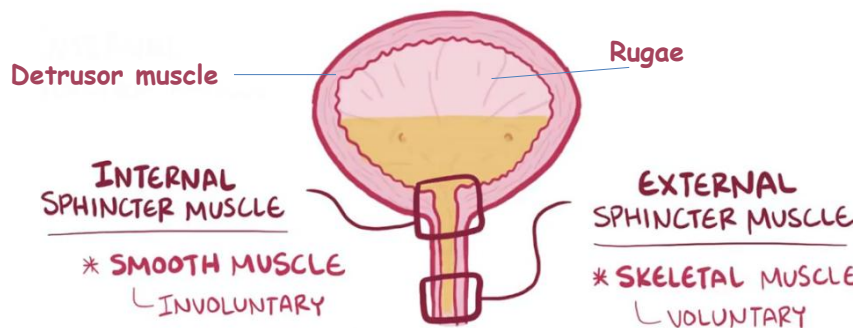
Observed in patient with chromosomal disorders like Turner syndrome, trisomy 21. Due to teratogens. Or Unknown

Histology

California Northstate University College of Dental Medicine

10

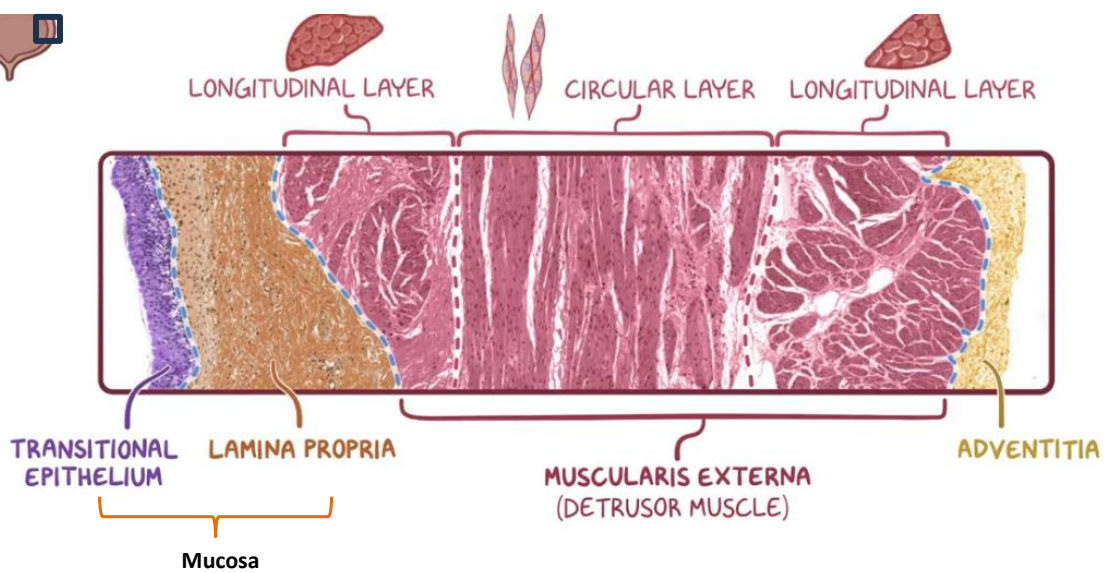
Bladder



Bladder: sac-shaped, temporary storage for urine (up to 750 ml of urine). Innervated by somatic (voluntary sphincter control), parasympathetic and sympathetic nerve fibers.

Composed of: body (connected w. ureter via ureterovesical junction) and neck (funnel-shaped connected w. urethra). Wall has rugae to allow its distention.

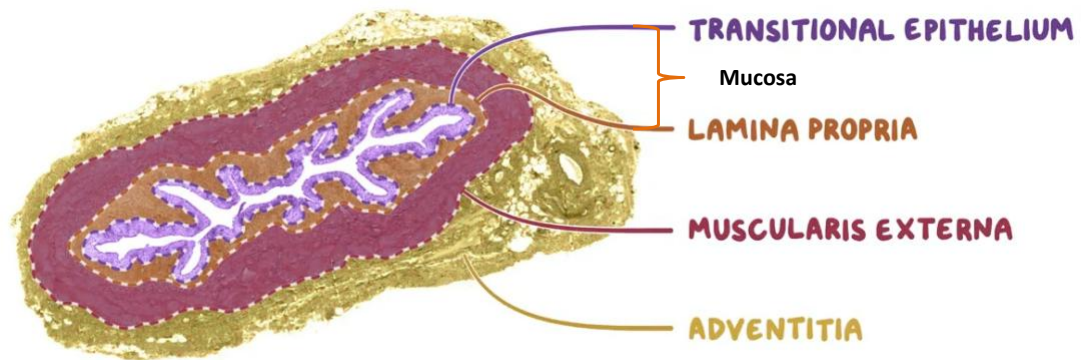
Detrusor muscle= smooth muscle of bladder. Sends sensory information (stretch receptors).



California Northstate University College of Dental Medicine

12

Ureters



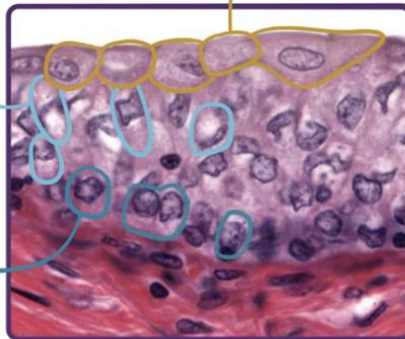
TRANSITIONAL EPITHELIUM

INTERMEDIATE LAYER

- ~ CUBOIDAL & COLUMNAR CELLS
- ~ DIFFERENTIATE to REPLACE DAMAGED UMBRELLA CELLS

BASAL LAYER

- ~ SINGLE LAYER of CUBOIDAL CELLS
- ~ REST on the BASEMENT MEMBRANE



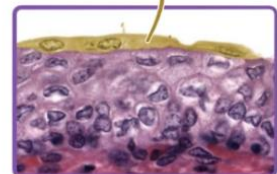
LARGE OVOID DOME/ UMBRELLA CELLS

- ~ CURVED APICAL SURFACE
- ~ COVER OTHER EPITHELIAL CELLS



WHEN the URETER is DISTENDED

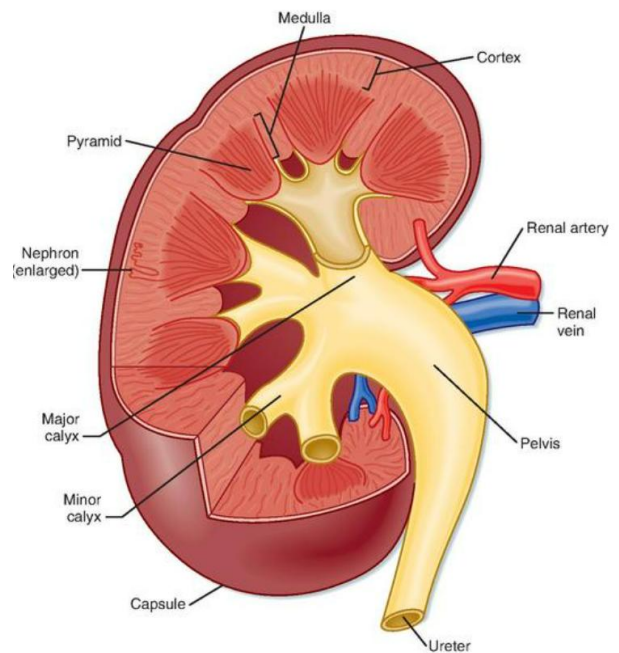
- ~ ↓ FOLDS in the EPITHELIUM
- ~ FLATTENING & ↓ OVERLAP of UMBRELLA CELLS



Transitional epithelium (urothelium) lines the ureter, bladder and proximal urethra. Expand to accommodate large volume of urine & impermeable to water/ions. Lined with umbrella cells.

Kidney Structure

- The kidney is composed of cortex, medulla & hilum. Nephron is the functional unit of the kidney.
- Cortex: outer region containing corpuscle and tubules.
- Hilum: Minor and major calyces. Penetrated by blood vessels, nerves, & ureter.
- Medulla: organized into renal pyramids. The renal pyramid drains into the minor calyx that drain to major calyx & ureter.



California Northstate University College of Dental Medicine

15

The kidney is surrounded by three layers of tissue (outside-inside):

1- Renal fascia (dense connective tissue), functions to anchor the kidney to its surroundings.

2- Adipose capsule (fatty layer): functions to protect the kidney from trauma.

3- Renal capsule (transparent fibrous connective tissue layer) with supportive and protective function.

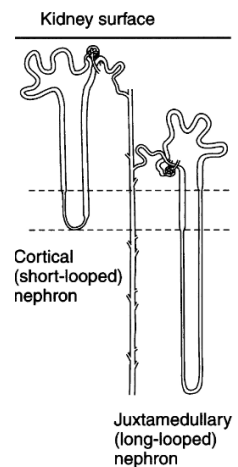
Types of nephrons

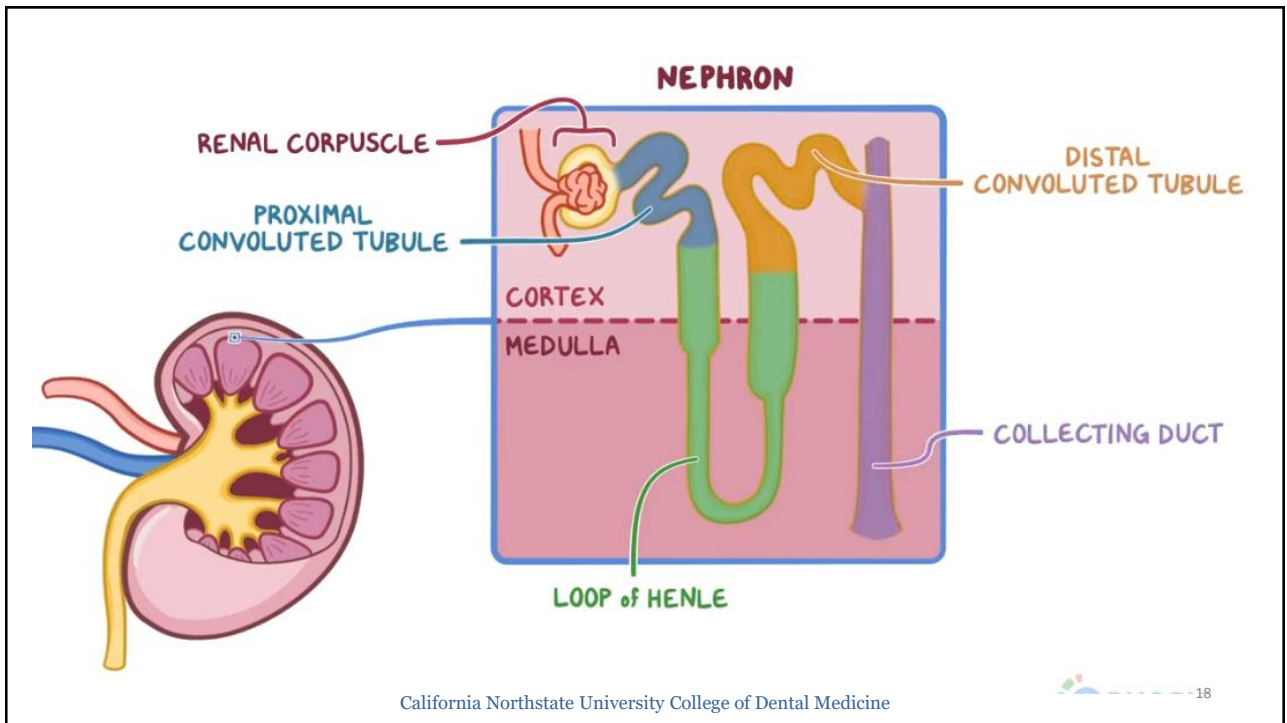
Cortical nephrons (85 % of the nephron population).

Relatively short loop of Henle. Involved in the excretory and regularity functions of the kidney.

Juxtamedullary nephrons (15 %). Long loop of Henle that extends deep into the renal medulla. Their corpuscle is in the corticomedullary border. Important for urine concentration and act as “salt conserving” nephrons.

Upon the reduction of blood volume, the human body directs a higher proportion of renal blood flow to the juxtamedullary nephrons to help conserve the extracellular fluid volume.





Renal corpuscle

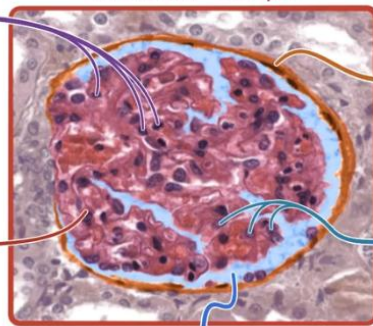
MESANGIAL CELLS

- * IRREGULAR SHAPED NUCLEI
- Involved in the immune defense and glomerular repair

GLOMERULUS

- * TUFT of CAPILLARIES
- * SUPPLIES BLOOD

FILTRATE/
ULTRAFILTRATE
~ NO BLOOD CELLS
or LARGE PROTEINS

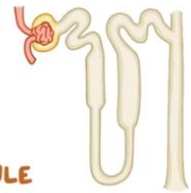


CAPSULAR/
BOWMAN'S SPACE

GLOMERULAR/ BOWMAN'S CAPSULE

- * OUTER PARIETAL LAYER
~ FLAT SIMPLE SQUAMOUS EPITHELIUM
- * INNER VISCERAL LAYER
~ COMPLEX EPITHELIAL CELLS called **PODOCYTES**
 - ↳ LARGE OVAL NUCLEI
 - ↳ WRAP AROUND EACH of the CAPILLARIES
 - ↳ form NARROW SLITS that FILTER BLOOD

ULTRAFILTRATE

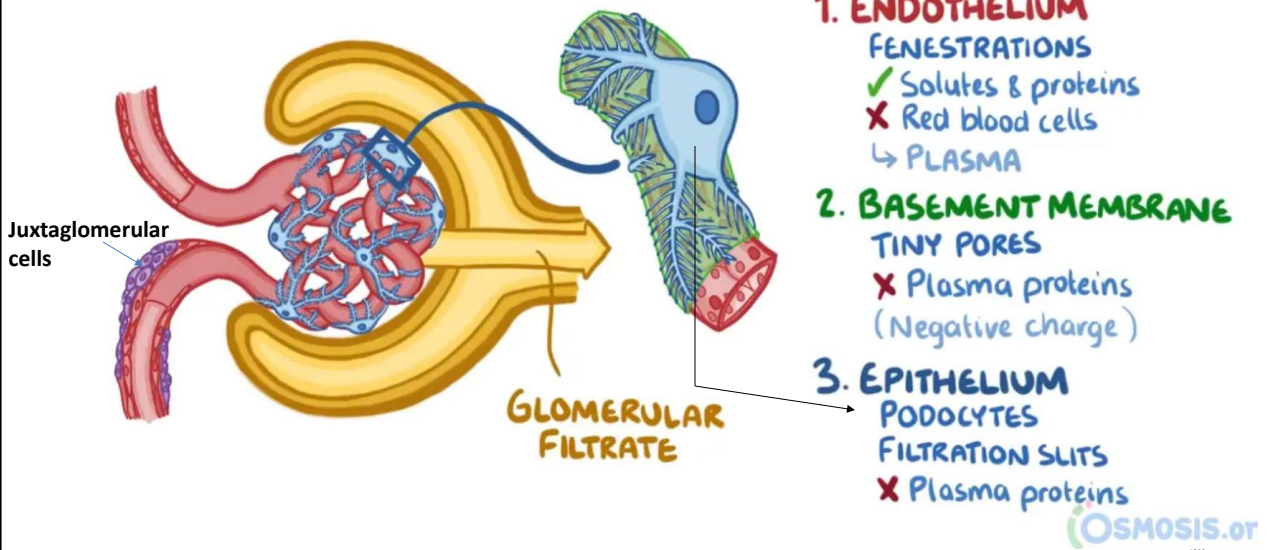


Renal corpuscle: composed of the glomerulus & Bowman's capsule. The glomerulus is composed of glomerular capillary network encased by the Bowman's capsule. The renal corpuscle filters 180 L of blood/day.

California Northstate University College of Dental Medicine

19

GLOMERULAR FILTRATION BARRIER - 3 LAYERS

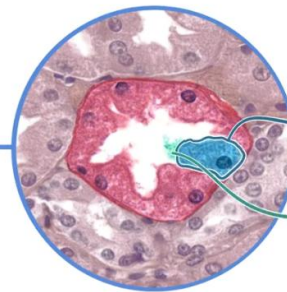
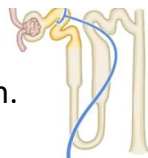


California Northstate University College of Dental Medicine

20

Proximal tubule

- Longest part of the nephron. Consists of convoluted and straight portions.
- Lined by simple cuboidal epithelial with microvilli to increase the surface area.



* REABSORBS ORGANIC NUTRIENTS, PROTEINS, WATER, & ELECTROLYTES

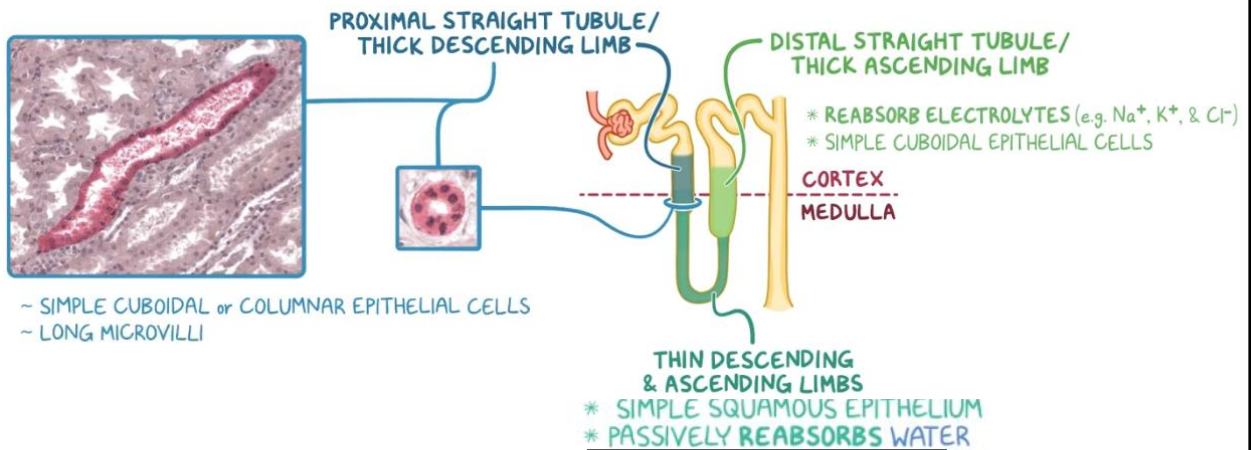
* SECRETES ANIONS & CATIONS (e.g. HYDROGEN & AMMONIUM)

SIMPLE CUBOIDAL or COLUMNAR EPITHELIAL CELLS

LONG MICROVILLI

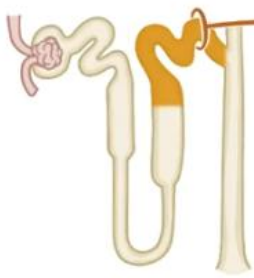
- Around 80 % of the glomerular filtrate is reabsorbed here. Reabsorbed material include glucose, amino acids, sod. & water (2/3 of total via aquaporin channels).
- Also secrete ions like hydrogen and ammonia.
- Glomerulus & proximal tubules are located in the kidney cortex.

Loop of Henle



In medulla, reabsorbs water (descending loop via aquaporin channel) and electrolytes (sodium, chloride and potassium from ascending part).

DISTAL CONVOLUTED TUBULE



* LIGHTER STAIN

* SHORTER MICROVILLI

* REABSORB Na^+ & Cl^-

↓
MAKES URINE
HYPOTONIC

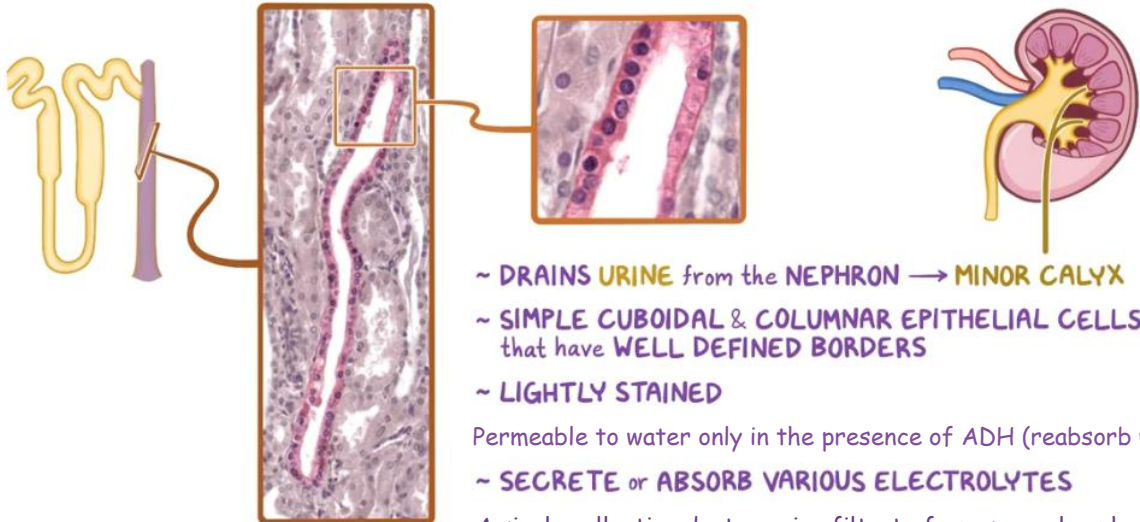
Maintain the extracellular fluid and electrolyte homeostasis.

- Selectively reabsorb electrolytes like sodium and chloride which renders the urine hypotonic.
- Secrete hydrogen and potassium ions.
- Can adjust the amount of water secreted in the urine according to stimulation by the anti-diuretic hormone.

California Northstate University College of Dental Medicine

23

COLLECTING DUCT



~ DRAINS URINE from the NEPHRON → MINOR CALYX

~ SIMPLE CUBOIDAL & COLUMNAR EPITHELIAL CELLS that have WELL DEFINED BORDERS

~ LIGHTLY STAINED

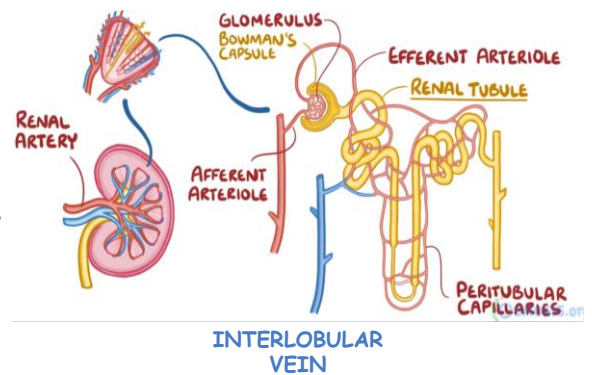
Permeable to water only in the presence of ADH (reabsorb water)

~ SECRETE or ABSORB VARIOUS ELECTROLYTES

A single collecting duct receive filtrate form several nephrons

Blood flow to the kidney

- Renal artery supplies the kidney with oxygenated blood. Branches to yield the afferent arterioles to the glomeruli.
- Next, blood drain into the efferent arterioles ➡ peritubular capillary network.
- Peritubular network facilitate water & ion exchange to occur along the length of renal tubules.
- The peritubular capillaries drain in the renal interlobular vein and converge to form the renal vein to inferior vena cava.



Regulation of renal blood flow

Vasodilation of afferent arterioles will increase GFR & RBF while vasodilation of efferent arterioles will decrease GFR.

- **Sympathetic stimulation:**
 - Kidney:
 - Decreases renal blood flow via vasoconstriction of afferent arterioles (via NE)
 - Stimulates renin release (leading to sodium reabsorption).
 - Can stimulate epinephrin release from the adrenal gland.
- **Parasympathetic** mainly affect the bladder.

- Others

- Angiotensin II (RAAS) and endothelin (released by damaged vascular endothelial cells of the kidneys) causes vasoconstriction & decrease the GFR.
- Nitric Oxide: Decreases renal vascular resistance (via vasodilation) and increases GFR. Basal nitric oxide levels are important for normal kidney function.
- Prostaglandins (PGE2 and PGI2) and Bradykinin: cause vasodilation afferent arteriole > efferent arteriole (decrease renal vascular resistance), increase renal blood flow, and increase GFR.

Kidney Function

California Northstate University College of Dental Medicine

28

Physiological Functions

1- Regulation:

- Water and electrolyte balance.
- Systemic blood pressure and extracellular fluid volume.
- Acid-base balance & calcium and phosphate balance.

2-Filtration:

- Excretion of metabolic waste and foreign substances.

3- Endocrine function:

- Renin (BP), Vitamin D activation and Erythropoietin (RBCs production).
- target organ for several hormones like ADH, angiotensin II, aldosterone, PTH..

4- Metabolic:

- Gluconeogenesis. Stimulated by epinephrine & inhibited by insulin.

Regulation of water and electrolyte Balance

Kidneys matches water and electrolyte output to the total input and body needs.

- Water intake:
 - Fluids ingested + metabolism generated around 2300 ml
- Water output:
 - Insensible water loss: skin & lungs, around 700 ml
 - Water loss by others (sweating, fecal and **urine**).

For example,

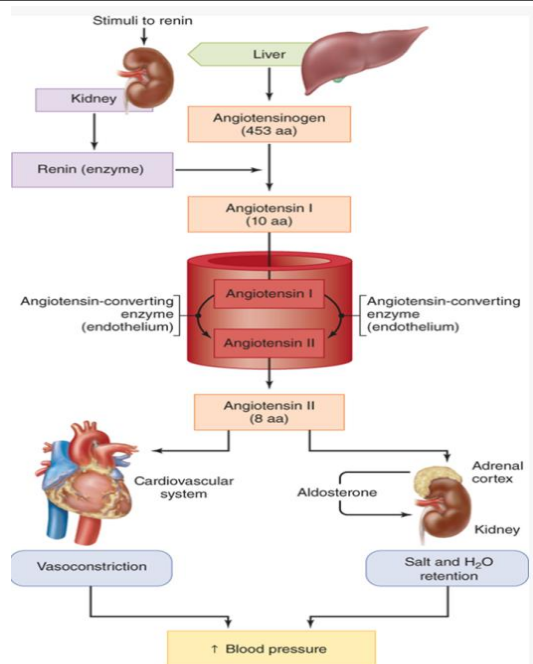
- Increased fluid intake decrease ADH secretion which decreases water reabsorption in the distal tubule and collecting ducts. Increase urine output (more dilute urine=lower osmolarity).
- Dehydration, diarrhea (water deprivation) increase ADH secretion and decrease urine volume (higher osmolarity).

Regulation of systemic blood pressure

- a) Regulate the ECF volume (blood volume) which impacts the venous return and CO.
- b) **Renin:** which activates the renin-angiotensin-aldosterone system (RAAS).
 - Hormonal mechanism for blood pressure regulation through vasoconstriction and fluid retention.
 - Influence peripheral vascular resistance & systemic arterial blood pressure.
 - Slower than the baroreceptor reflex.
 - Released by the juxtaglomerular cell of the kidney. Activated in conditions that decrease BP or blood volume.

Components: Renin (enzyme, kidney), angiotensinogen (from liver), ACE (enzyme, many places, high conc in lung), aldosterone (mineralocorticoid, adrenal gland).

Stimuli: Hypotension or sodium depletion or sympathetic stimulation induce the release of renin from the juxtaglomerular cells and initiate the RAAS mechanism.



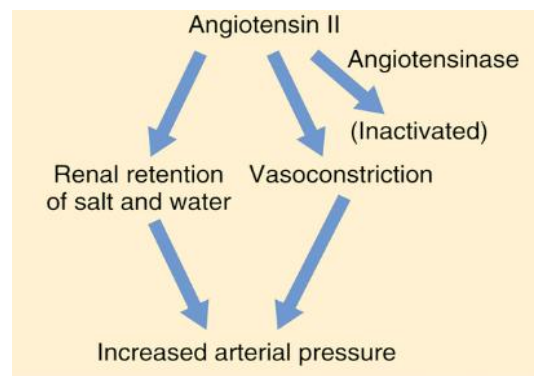
California Northstate University College of

Source: Adel Elmosehiti:
Cardiology: An Integrated Approach
Copyright © McGraw-Hill Education. All rights reserved.

33

Angiotensin II:

- Remains in the blood for 1-2 minutes (inactivated by angiotensinases).
- Peptide with potent vasoconstriction properties. Vasoconstriction occurs more in the arterioles >> veins. Constriction of the arterioles increases the total peripheral resistance and bp.
- Ag II increase sodium reabsorption from the proximal convoluted tubule (via increases $\text{Na}^+ - \text{H}^+$ exchange).



- Ag II stimulates the secretion of aldosterone from adrenal cortex.
 - Aldosterone works on the distal and collecting tubules. It directly increase Sod. reabsorption which will help the kidney retain water. Collectively, increase the blood volume and BP.
- Ag II can also affect hypothalamus to increase thirst and induce vasopressin release.

As a protective mechanism, the endothelial cells release nitric oxide to protect against excessive vasoconstriction.

Vasopressin (antidiuretic hormone, ADH): peptide synthesized in the hypothalamus and secreted from the pituitary gland in response to many stimuli.

Causes:

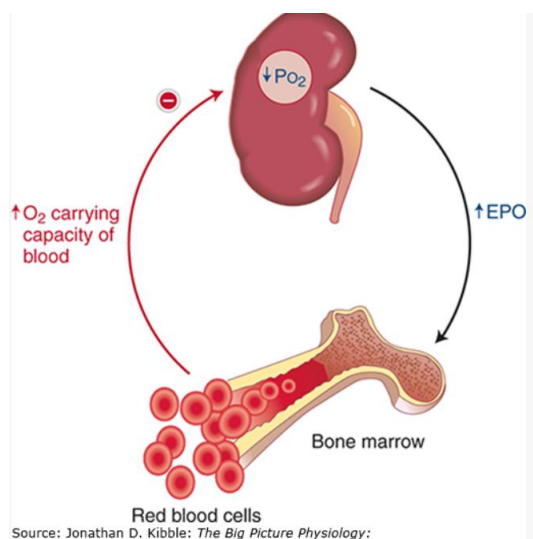
- 1- Vasoconstrictor for vascular smooth muscle.
- 2- Increasing the water reabsorption in distal convoluted tubule and collecting duct via increasing the aquaporin -2 channels to increase blood volume. Excess ADH secretion can cause oliguria (<400 mL/day).

Secretion of Erythropoietin

In response to sensing changes in oxygenation levels, Renal Epo-producing cells secrete erythropoietin.

Erythropoietin (Epo) stimulates RBCs production from the bone marrow.

Renal failure can cause anemia.

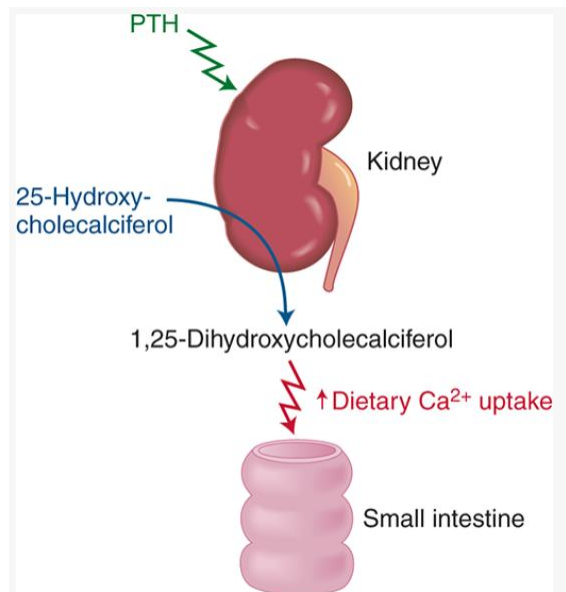


Source: Jonathan D. Kibble: *The Big Picture Physiology*:

Activation of vitamin D.

Following initial metabolism in the liver, the Kidneys convert the inactive form of vitamin D to the active calcitriol (1,25dihydroxycholecalciferol) under the influence of parathyroid hormone (PTH).

Active form is needed for calcium absorption.



Source: Jonathan D. Kibble: *The Big Picture Physiology*:

Regulation of calcium and phosphate balance

Kidneys and GI share the responsibility to keep the calcium and phosphate balance to preserve the bone integrity.

Kidneys initially filters magnesium, calcium and phosphate through glomerulus. In tubules, they reabsorbed to maintain homeostasis (hormonal influence).

For example, parathyroid hormone increases calcium reabsorption but inhibit phosphate reabsorption. Therefore, PTH causes phosphaturia.

Excretion of metabolic waste and foreign substances.

Metabolic end products (urea, uric acid, creatinine, urobilin and others), collectively named uremic toxins. Renal failure can lead to “**uremia**”. Uremia is associated with nausea, vomiting, skin itching, fatigue and weight loss.

Kidney Terminology

Term	Definition
Filtration	The process by which water and solutes in the blood leave the vascular system through a filtration barrier and enter the Bowman's space.
Secretion	Refers to the transport of secreted substances from the cytosol of epithelial into the tubular lumen.
Reabsorption	Involves the reabsorption of molecules from the filtrate (urine) and returning it back into the blood.
Excretion	Refers to substances present in the final urine produced by the kidneys and ready to exit the body

- **Renal Blood Flow:**

- volume of plasma delivered to the kidneys per unit time.
- 600 mL/min in a normal adult
- Measured by para-aminohippuric acid (PAH) clearance
- Reflects renal hemodynamics and tubular function

- **GFR**

- Volume of plasma filtered by all glomeruli per unit time.
- 125 ml/min in adult
- Measured by inulin or creatinine clearance
- Reflect filtration capacity

Basic nephron function

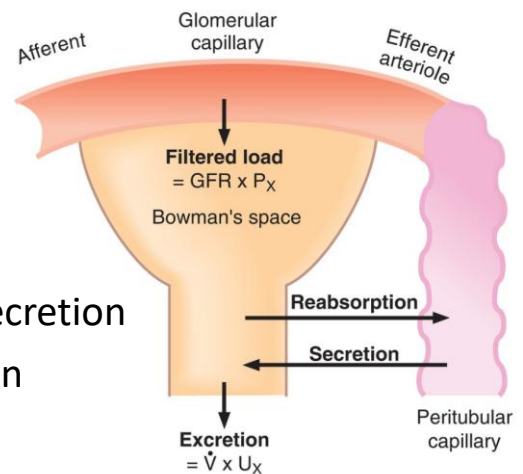
Excretion = Filtration – Reabsorption + Secretion

Filtered load = GFR x plasma concentration

Excretion (mg/min) = $V \times U_x$

Clearance (ml/min) = $(V \times U_x) / P_x$

V , urine flow rate (V); U_x = Urine concentration of solute x ; P_x = Plasma concentration of solute x



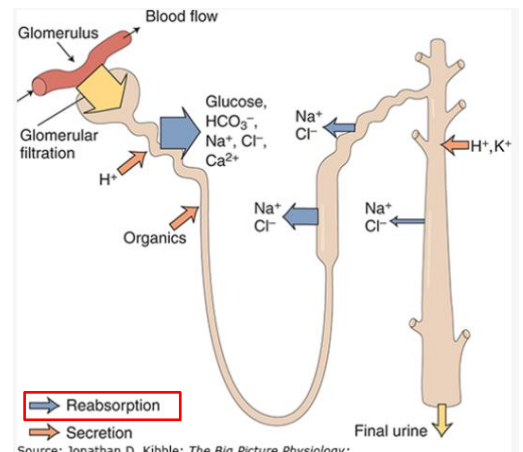
Glomerular filtration

- Plasma volume is filtered by the kidneys around 60 times per day (125ml/min).
- Filtrate collects in the Bowman's capsule and passes to the tubules for further manipulation.
- Filterability is inversely proportional to the molecule size.
- Glomerular filtration step mainly acts like a mesh. Thus, tubular reabsorption and secretion are important to regulate the amount of excreted water and electrolytes.

Filtered	Not filtered
Water, electrolytes, low molecular weight proteins and small peptides.	Fatty acids, large size plasma proteins as albumin and globulin.

Tubular reabsorption

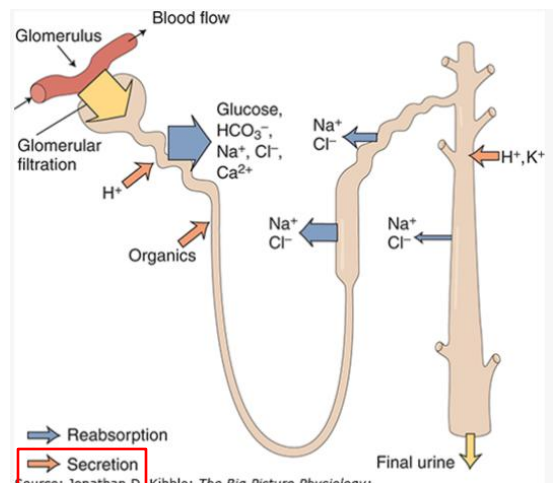
- The filtrate (tubular fluid) flows from the Bowman's capsule through proximal tubule, loop and distal tubules.
- Reabsorption occurs by simple diffusion or via active transport using specific transporters.
- Reabsorption of useful components is either complete (like, glucose), or almost complete (water and electrolytes).
- Reabsorption of waste products is minimal (like, urea).



Renal Physiology and Acid-Base Balance | The Big Picture Physiology: Medical Course & Step 1 Review, 2e | AccessMedicine | McGraw Hill Medical (cnsu.edu)

Tubular secretion

- Tubular secretion defines the secretion of ions like hydrogen, potassium or ammonia or drug metabolites into the tubules and it is excreted by active transport or passive transport.
- Hydrogen ion is secreted to facilitate the maintenance of the plasma pH.



Hormonal regulation

- **Angiotensin II and Aldosterone**, increase NaCl reabsorption and increase hydrogen secretion. This will indirectly increase water reabsorption.
- **ADH**, increase water reabsorption in the distal and collecting tubules.
- **Arterial natriuretic peptide (ANP)** decreases NaCl (& water) reabsorption in distal and collecting tubules to respond to increased atrial blood pressure.
- **PTH**, increases calcium reabsorption and decreases phosphate reabsorption.

How does Kidneys Conserve Water?

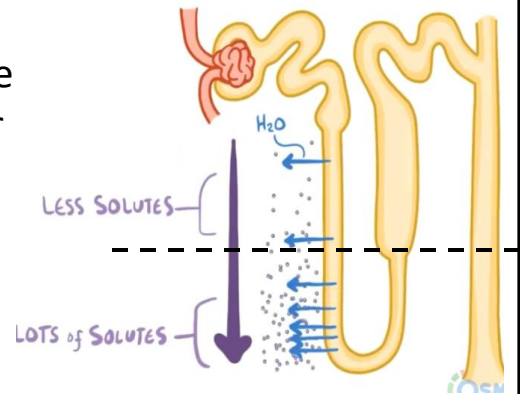
By Excreting Concentrated Urine

California Northstate University College of Dental Medicine

48

Corticopapillary gradient

- This is the gradient of osmolarity from the cortex (300 mOsm/L) to the papilla [inner medulla] (1200 mOsm/L).
- This osmolarity gradient is maintained by electrolytes and urea in the interstitium.
 - Electrolytes (Sodium, Potassium, chloride) establish the countercurrent multiplier mechanism.
 - Urea establishes urea recycling.

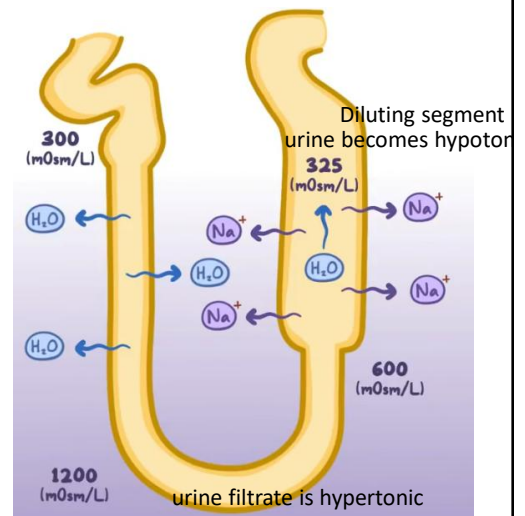


Countercurrent Multiplication Mechanism

In loop of Henle.

The nephron uses electrolyte pump & energy to generate an osmotic gradient that enables water reabsorption (descending L) & salts (ascending L) to produce more concentrated urine with the correct osmolarity.

Urine filtrates starts as isotonic solution in the glomerulus, then changes to hypertonic followed by hypotonic solution in the loop of Henle and distal tubule.

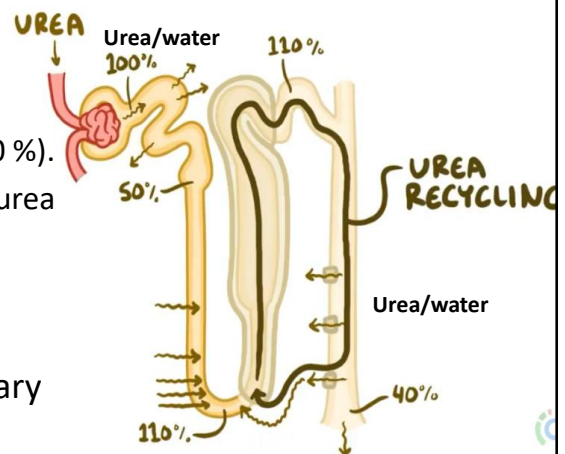


Urea Recycling

Urea is the byproduct of protein metabolism.

- 1- Freely filtered from the glomerulus.
- 2- Partly reabsorbed from the PCT (50 %).
- 3- Secreted back into descending loop of Henle (110 %).
- 4- Reabsorbed in DCT /collecting duct. Reabsorbed urea goes back to the loop of Henle "Urea recycling".
- 5- 40 % urea load is excreted in urine.

Urea recycling help establish the corticopapillary gradient necessary to ensure maximum water conservation.



Urine Flow

Collecting ducts → Renal papilla → minor calyx → major calyx → renal pelvis
→ ureter → urinary bladder → urethra.

Ureters are inserted in the bladder with oblique orientation to create one-way valve and prevent the backflow of urine.

Urine is stored in bladder till micturition.

(MILLIONS of NEPHRONS MAKE URINE)

URINE FLOW:

MINOR CALYX

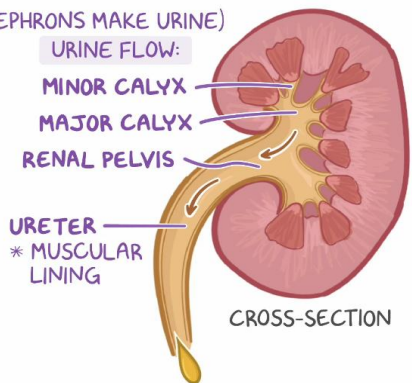
MAJOR CALYX

RENAL PELVIS

URETER

* MUSCULAR
LINING

CROSS-SECTION



Micturition

- Micturition: process whereby the urinary bladder empties when it becomes filled.
- Micturition reflex is an autonomic spinal cord reflex, but it is modulated by centers in the cerebral cortex. Spinal cord injury can lead to urinary incontinence.

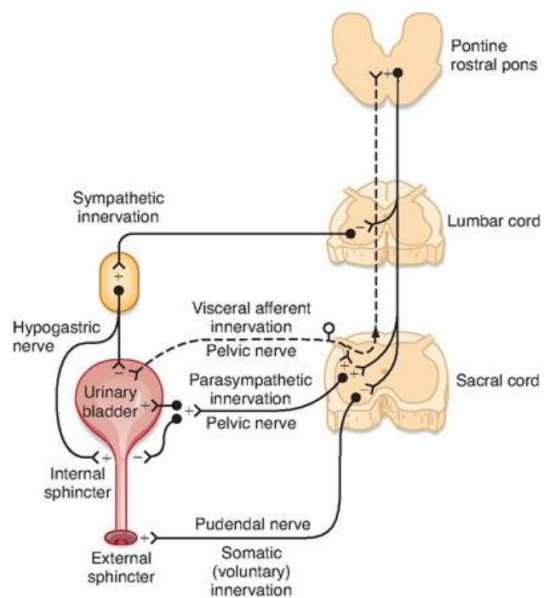
Two main steps:

- 1- Bladder fills until tension > threshold level, sensed by mechanoreceptors.
- 2- Micturition reflex: nervous reflex that inhibit sympathetic and activate parasympathetic pathways. Leads to micturition or conscious urgency to urinate.

Sympathetic: sympathetic axons inhibit detrusor muscle contraction and constrict internal sphincter (via NE, inhibit micturition).

Parasympathetic stimulating bladder (detrusor muscle) contraction and relax the internal sphincter (via Ach, promote micturition).

Somatic innervation to external sphincter.



Composition of Urine

- Clear, yellowish color, 95 % water and 5 % solutes
- Solutes include:
 - 1- Nitrogenous waste like urea, uric acid and creatinine.
 - 2- Electrolytes like sodium, potassium, calcium.
 - 3- Others
- pH around 6 (range 5-7).
- Volume: 0.8-2 L/day.

A normal 70-kg person must excrete 600 ml of urine per day to eliminate metabolic waste products.

Amount of water in the final urine is controlled by ADH and water intake.

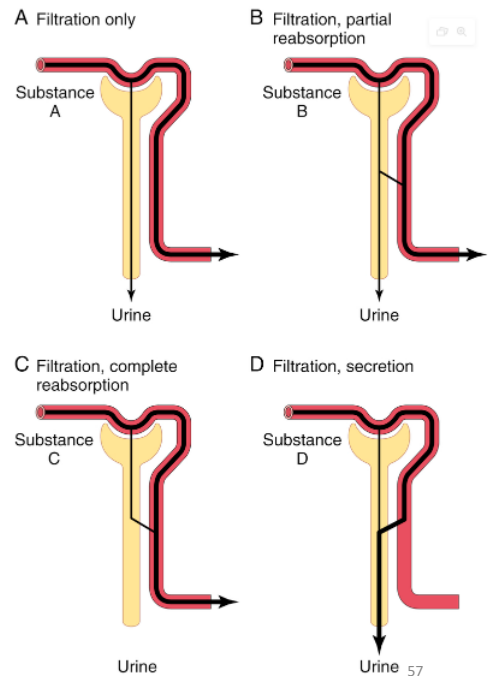
Substance	Concentration
Na ⁺	50–130 mEq/L
K ⁺	20–70 mEq/L
Ammonium (NH ₄ ⁺)	30–50 mEq/L
Ca ⁺⁺	5–12 mEq/L
Mg ⁺⁺	2–18 mEq/L
Cl ⁻	50–130 mEq/L
Inorganic phosphate (Pi)	20–40 mEq/L
Urea	200–400 mmol/L
Creatinine	6–20 mmol/L
pH	5.0–7.0
Osmolality	500–800 mOsm/kg H ₂ O
Glucose	0
Amino acids	0
Protein	0
Blood	0
Ketones	0
Leukocytes	0
Bilirubin	0

Renal Handling for Substances

California Northstate University College of Dental Medicine

56

- A) Filtered only not reabsorbed or secreted (for example creatinine)
- B) Filtered and partial reabsorption (common scenario).
- C) Filtered and complete reabsorption (glucose and amino acids)
- D) Freely filtered and additional amount are secreted (organic acids/bases and some drug metabolites).



California Northstate University College of Dental Medicine

Reabsorption of Glucose

- Filtered glucose is 100% reabsorbed from filtrate via SGLT-2 transporter in the proximal tubules.
- Glucosuria is presence of glucose in urine. It is associated with plasma glucose level >180 or 200 mg/dl due to oversaturation of the carrier system leading to urinary excretion of glucose.
- Chronic hyperglycemia is a known risk factor for chronic kidney disease.
- Excess glucose glyicates plasma proteins via non-enzymatic glycation mechanism which eventually lead to glomerular damage and declining kidney function.

Reabsorption of Water

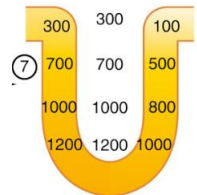
- Water is freely filtered 180 L/day.
- Mainly reabsorbed from the proximal tubule (65 %) via simple diffusion (w. sodium)
- Countercurrent mechanism and descending loop of Henle take part in water reabsorption.
- ADH (w. urea recycling) complete the water reabsorption in the distal tubule and collecting duct (up to 99%).
- Excreted amount is around 1.5 L/day

Disorders of Urinary Concentrating Ability

Impairment in the kidneys' ability to concentrate or dilute the urine can occur due to:

- 1. Inappropriate secretion of ADH:** leads to abnormal water excretion. For example, head injuries, tumors, infections or congenital factors can decrease the release ADH from the posterior pituitary (central diabetes insipidus).
- 2. Nephrogenic Diabetes insipidus:** Inability of distal & collecting tubules and collecting ducts to respond to ADH.
- 3. Impairment of countercurrent mechanism** due failure to establish hyperosmotic medullary interstitium.

Even if adequate ADH is present, maximal urine concentration is limited by the degree of hyperosmolarity of the medullary interstitium.



Electrolytes

- Most unbound electrolytes are freely filtered through the glomerulus.
- Around 2/3 of the filtered load is reabsorbed in the PCT (calcium, phosphate, sodium and potassium).
- Majority of filtered magnesium (60 %) , and majority of remaining sodium, calcium & potassium (≥ 20 %) are re-absorbed from the thick ascending loop of Henle. Levels affected by loop diuretics
- In the PCT, sodium acts as a co-transporter that facilitates glucose, phosphate and amino acids reabsorption. Sodium also facilitates water reabsorption.

Kidney function tests

- Urine examination
- Renal clearance.
- Glomerular filtration rate.
- Renal plasma flow.
- Filtration fraction (FF) = GFR: RPF ratio [0.15-0.20 (15-20%)].
- Blood urea nitrogen (BUN).

Urine examination

1- Gross assessment (color, clarity, odor). Turbid urine can indicate the presence of uric acid, calcium oxalate, or phosphate crystals, blood cells or protein. Protein in the urine creates foaming when shaken. Yellow or orange foam can be indicative of bile pigments.

2- Urine dipstick analysis provides semiquantitative assessment for osmolality, pH, presence of heme, albumin or glucose.

3- Microscopic examination of urine sediment provides a tool to screen for casts, presence of cells (WBCs or RBCs), crystals, bacterial infection or lipid droplets. WBCs in the urine (pyuria) are indicative of urinary tract infection.

Renal clearance

- Renal clearance is the volume of plasma freed of a given substance/min.
- $\text{Clearance} = (\text{urine concentration} * \text{urine flow rate}) / \text{plasma concentration}$.
- It reflects the ability of the kidney to clear metabolic end products and toxic metabolites. Does not account for the reabsorption or secretion processes.
- Renal clearance uses urine sample collected over 24 h using inulin or creatinine as a reference.

Clearance ratio

Why inulin?

Inulin: physiologically inert substance that is freely filtered at the glomerulus, and is neither secreted, reabsorbed, synthesized, nor metabolized by the kidney.

Clearance ratio = clearance of substance/clearance of inulin

- Clearance ratio =1 “substance is filtered but not secreted or reabsorbed.
- Clearance ratio > 1, substance is filtered and actively secreted by the kidney.
- Clearance ratio <1, substance is not filtered (albumin) or filtered but reabsorbed (for example, glucose).

Glomerular Filtration Rate (GFR)

- GFR = sum of filtration rates of all functioning nephrons.
 - Can be detected with an endogenous reference (creatinine) or exogenous reference (inulin).
 - A decline in the GFR usually indicates kidney diseases or decreased kidney perfusion. GFR slightly decline with aging.

A) Measured GFR (mGFR)

Detected with the exogenous reference (inulin).

B) Estimated GFR (eGFR)

Estimated with an updated mathematical equation that uses serum creatinine level and accounts for age, gender and weight.

Why creatinine?

- Creatinine is derived from creatine metabolism in skeletal muscles.
- Released into the circulation at a relatively constant rate.
- Creatinine is freely filtered across the glomerulus. Not reabsorbed or metabolized. Thus, can be used as an endogenous control after normalization to muscle mass.
- However, creatinine levels can be affected by other factors like diet; muscle mass, malnutrition and other chronic illnesses.

Inulin

- Inulin is an exogenous reference used to detect the measured GFR. The amount of inulin filtered must equal the amount of inulin excreted.

CKD-EPI Creatinine Equation (2021)

Expressed as a single equation:

$$eGFR_{Cr} = 142 \times \min(S_{Cr}/\kappa, 1)^\alpha \times \max(S_{Cr}/\kappa, 1)^{-1.200} \times 0.9938^{Age} \times 1.012 \text{ [if female]}$$

where:

S_{Cr} = serum creatinine in mg/dL

κ = 0.7 (females) or 0.9 (males)

α = -0.241 (female) or -0.302 (male)

$\min(S_{Cr}/\kappa, 1)$ is the minimum of S_{Cr}/κ or 1.0

$\max(S_{Cr}/\kappa, 1)$ is the maximum of S_{Cr}/κ or 1.0

Age (years)

GFR & stages of kidney disease

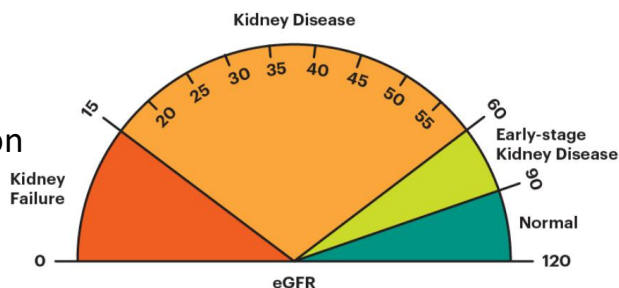
Stage 1 - Normal kidney function (GFR >90 ml/min/1.73 m²)

Stage 2 - Mild loss of kidney function (GFR= 60 – 90 ml/min/1.73 m²)

Stage 3 - Moderate loss of kidney function (GFR=30 – 60 ml/min/1.73 m²)

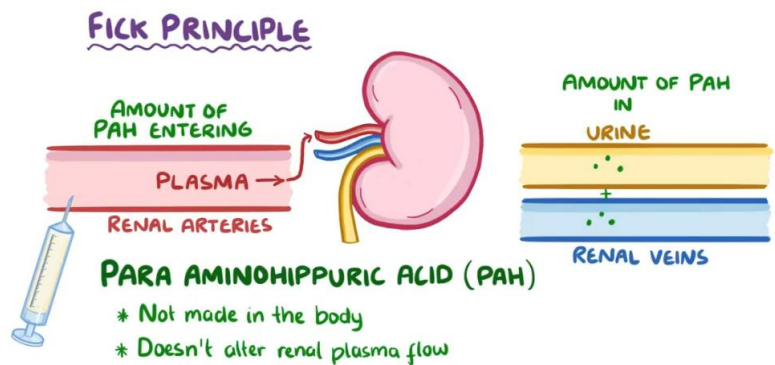
Stage 4 - Severe loss of kidney function (GFR = 15 – 30 ml/min/1.73 m²)

Stage 5 - End-stage renal disease (ESRD, GFR < 15 ml/min/1.73 m²)



Renal plasma flow

The amount of a substance in the blood that flows into an organ equals the amount that flows out of that organ. PAH is bound to plasma protein. Therefore, PAH is secreted by the proximal tubules and not filtered or reabsorbed.



Therefore, infusing PAH and comparing its urine and plasma concentrations can indicate the blood flow to the kidney and kidney function.

Blood Urea Nitrogen

- Urea is the product of protein catabolism. Freely filtered in the glomerulus.
- BUN refers to the serum concentration of urea.
- BUN level is inversely proportional to the GFR.
- Used as an indicator for kidney function. Normal level 7 - 24 mg/dL.
- Azotemia is the elevation of nitrogenous waste (like urea and creatinine) in the blood.
- Increased nitrogenous waste is indicative of a decreased GFR (acute and chronic kidney injury).

Regulation of ACID/BASE Balance

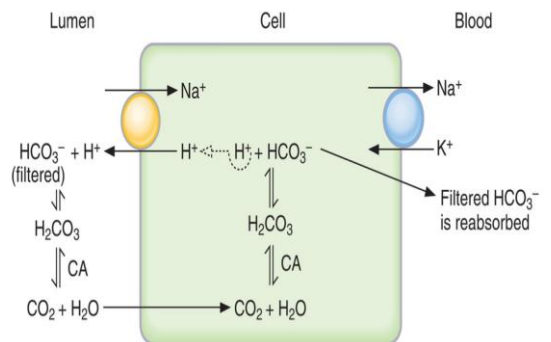
California Northstate University College of Dental Medicine

72

Regulate acid/base balance

To maintain the acid/base homeostasis (pH 7.45)

- Renal excretion is the first line to remove hydrogen ions from the human body. Further, the kidneys can also excrete non-volatile acids which are produced in the body including lactic acid, pyruvic acid and phosphoric acid.
- Maintain plasma bicarbonate [HCO_3^-] within the range of 22–28 mmol/L.



Excretion of hydrogen on filtered phosphate/ammonia

Kidneys help regulate blood pH by excreting hydrogen ion.

To prevent the decrease of urine pH by H^+ secretion, the kidney use chemical buffers (ammonia or phosphate buffer system) to enable hydrogen ion excretion.

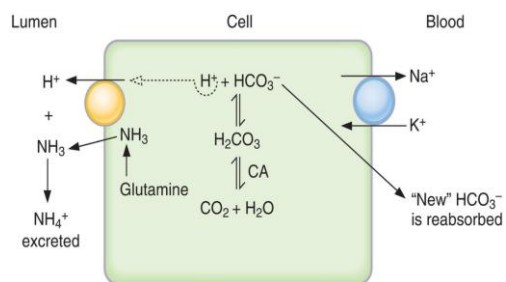


FIGURE 5.23 Mechanism for excretion of H^+ as NH_4^+ . CA = carbonic anhydrase.

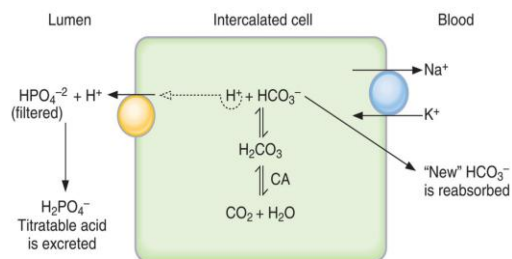


FIGURE 5.22 Mechanism for excretion of H^+ as titratable acid. CA = carbonic anhydrase.

Renal compensation in acid-base disorders

A) Acidosis:

- 1- Metabolic acidosis due to increased acid production (like ketoacidosis associated with diabetes) and renal failure.
- 2- Respiratory acidosis due to hypoventilation (opiates or sedative overdose, COPD).

Renal compensation consists of increased excretion of H^+ and increased reabsorption of HCO_3^- .

B) Alkalosis:

1- Metabolic alkalosis due to vomiting, diuretics toxicity.

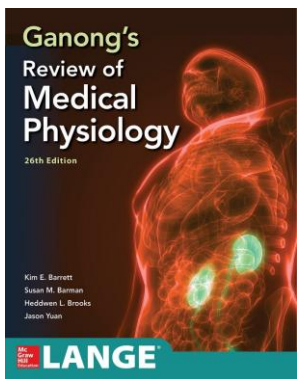
Renal compensation consists of increased excretion of HCO_3^- .

2- Respiratory alkalosis due to hyperventilation.

Renal compensation consists of decreased excretion of H^+ and decreased reabsorption of HCO_3^- (decrease serum bicarbonate level).

In respiratory acidosis and alkalosis, renal compensation occur in chronic conditions.

Resources



Topics ▾



Organ Systems

Renal system ▾